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Review

Liquid electrolyte: The nexus of practical lithium metal batteries

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SUMMARY

The specific energy of commercial lithium (Li)-ion batteries is reaching the theoretical limit. Future consumer electronics and electric vehicle markets call for the development of high energy density Li metal batteries, which have been plagued by poor cyclability. Electrolyte engineering can afford a promising approach to address the issues associated with Li metal batteries and has recently resulted in much improved cycle life under practical conditions. However, gaps still exist between the performance of current Li metal batteries and those required for commercial applications. Further improvements will require systematic analysis of existing electrolyte design methodologies. In this review, we first summarize recent approaches of advanced electrolytes for Li metal batteries paired with high-voltage cathodes. We then extract common features among these advanced electrolytes and finally discuss the future rational design directions and strategies.

INTRODUCTION

Lithium (Li)-ion batteries (LIBs) have dramatically changed our society with their broad applications in portable electronics and electric vehicles. However, their specific energies are reaching the theoretical limit with their graphite-based anodes.[1] Substantial efforts in both industry and academia have been made to revive Li metal batteries (LMBs), a technology explored and commercialized before LIBs but was prevented from widespread adoption due to its safety concerns and limited cyclability.[2] With the highest specific capacity (3,860 mAh/g) and the lowest reduction potential (−3.04 V versus standard hydrogen electrode [SHE]), Li metal anodes (LMAs) can potentially enable batteries with the highest specific energy.[3,4] However, these batteries usually have shorter cycle lives and potential safety hazards due to the high reactivity and volume fluctuations of LMAs.[5,6] Multiple strategies have been explored to address these issues, including 3-dimentional "host" structures,[7–13] artificial coatings,[14–22] and electrolyte engineering.[6,23–27] However, further developments are still needed for LMBs to meet the requirements of practical applications.

Ether-based electrolytes were predominantly used in the early decades of LMA research due to their decent reductive stability.[28–30] However, with the discovery of high-voltage cathodes, such as layered transition metal oxides $LiNi_xMn_yCo_zO_2$ (NMC),[31] common ether solvents fell short in Li metal full cells due to their poor oxidative stability.[32] It is noteworthy that we herein describe >4.2 V as high voltage, because this upper voltage cut-off is commonly used in both academia and industry, covering NMC cathode family, $LiCoO_2$ (LCO), $LiMn_2O_4$ (LMO), $LiNi_{0.5}Mn_{1.5}O_4$ (LNMO), $LiCoPO_4$ (LCP), Li-rich cathodes, etc. With the successful commercialization

Context & scale

Lithium (Li)-ion batteries have significantly advanced our society with their broad applications in portable electronic devices, electric vehicles, and grid storage. However, the energy density of Li-ion battery systems is reaching the theoretical limit, therefore, raising the urgent need for further improvement in the energy density of next-generation battery systems. Li metal batteries have been acknowledged as a promising alternative choice, yet they have been plagued by poor cyclability, which originates from significant electrode volume change, dendritic growth, and uncontrollable side reactions during Li metal cycling.

During the past decades, cycling efficiency of Li metal anodes have been significantly improved from ~80% to ~99.5% through gradual exploration and optimization of electrolyte formulations. Although still insufficient for commercial applications, such progress demonstrates the feasibility of further electrolyte engineering in enabling Li metal batteries. In this review, we systematically summarize past designs of Li metal battery electrolytes, conclude the key features of advanced electrolyte formulations, and then propose detailed design principles and methodologies that are critical for future developments of advanced Li metal battery electrolytes.

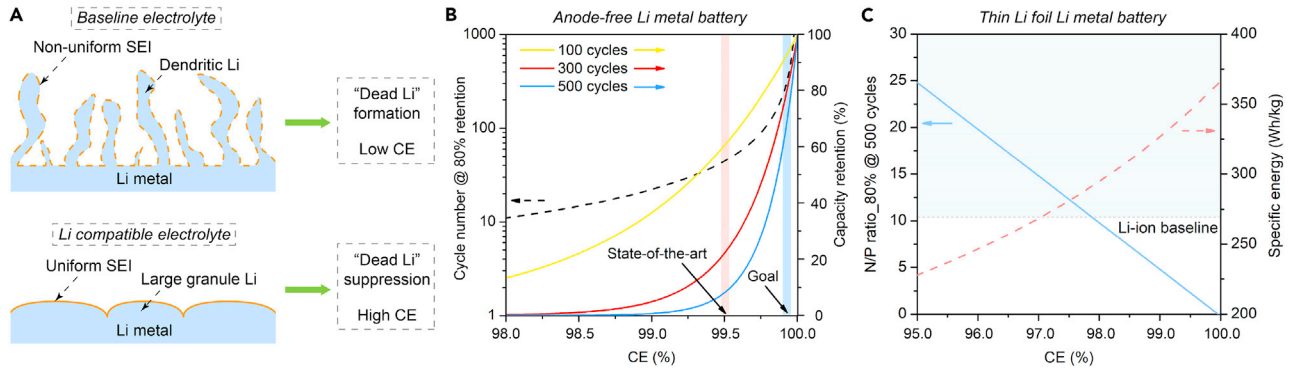


Figure 1. Opportunities and challenges of electrolyte engineering for LMBs

(A) Schematic comparing Li metal cycling in baseline electrolyte and in recently developed high-performance electrolytes.

(B) Theoretical plot of cycle number after reaching 80% capacity retention and capacity retention after 100, 300 and 500 cycles as a function of CE for anode-free LMBs.

(C) Theoretical plot of the N/P ratio required to achieve 80% capacity retention after 500 cycles and the corresponding full-cell specific energy as a function of CE for thin Li foil LMBs.

of LIBs, carbonate-based, standard-concentration electrolytes were also investigated for the use in LMBs.[7,9,33] Although these electrolytes were compatible with LIBs using high-voltage cathodes, they have been found unstable for Li metal cycling. They form non-uniform solid-electrolyte interphase (SEI), leading to dendritic growth, extensive "dead Li" formation,[2] low Coulombic efficiency (CE), and poor cycle life (Figure 1A). In the last decade, researchers started to discover new strategies, including enhancing electrolyte concentrations,[34] designing new fluorinated solvent molecules,[35] tuning electrolyte additives,[36] etc., that dramatically improved the electrolyte stability for use with both LMAs and high-voltage cathodes. These new electrolytes enabled more stable and uniform SEIs, larger granules of deposited Li, suppressed "dead Li" formation, and the subsequent high CEs[37,38] (Figure 1A). Recently, the cycle life of LMBs has been raised to an unprecedented level simply by tuning the electrolyte formulation. Therefore, liquid electrolyte engineering is a promising approach to enabling practical LMBs.[6]

Despite these successes, a considerable gap still exists between current LMB performance and practical requirements when taking specific energy and cycle life as the primary figure of merit.[39] For example, for an anode-free LMB to achieve 80% capacity retention after 500 cycles, a Li metal cycling CE of >99.96% is needed (Figure 1B). With the intrinsically impeded lithiation at the end of the 1st discharge of NMC cathodes,[40,41] ~10%–15% of Li remains on the anode side serving as an excess Li reservoir after the first cycle, but still a CE of >99.93% is needed. Currently, state-of-the-art LMB electrolytes only achieved CEs around ~99.5%.[35,42] LMBs with thin Li foil anodes can ease the requirements on CE, but the additional Li reservoir can lower the specific energy of the battery. To simultaneously achieve 350 Wh/kg and 80% capacity retention after 500 cycles, a CE of >99.62% is needed with a N/P ratio (the ratio of negative electrode capacity to positive electrode capacity) of 1.7 (Figure 1C). Although some electrolytes are approaching this CE value for LMAs, realizing the corresponding full cell with lean electrolyte and limited Li excess is still challenging[43] because of failure mechanisms other than the loss of active Li, such as electrolyte dry-out or impedance build-up.[44] Furthermore, the use of thin Li foil increases the cost and complexity of large-scale cell production,[39] adding concerns to its commercialization.[45] In summary, innovations on LMB electrolytes are needed to further improve the CE of LMAs and enable the successful development of LMBs. To guide

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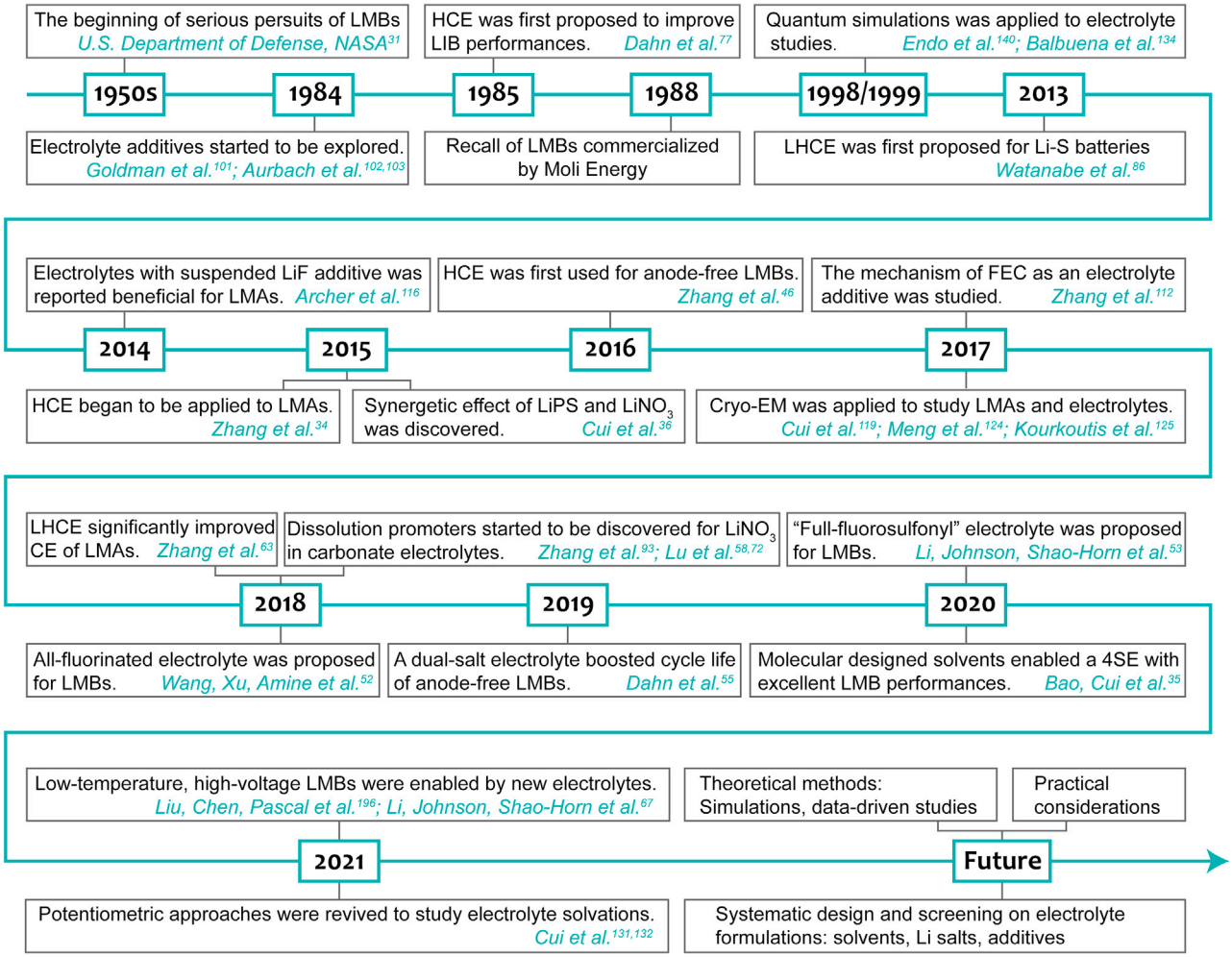


Figure 2. A roadmap of electrolyte developments for LMBs

further design and optimization of electrolyte formulations, systematic reflection of recent electrolyte engineering research is necessary.

In this review (Figure 2; Tables 1 and 2; Appendix 1), we first summarize recent electrolyte design strategies for LMBs with high-voltage cathodes. Then, we identify several shared features and properties of these new electrolyte formulations that are potentially the key directions to enable long cycle life of LMBs. Finally, we propose several strategies to further optimize the electrolyte formulations for practical LMBs.

RECENT PROGRESS IN HIGH-PERFORMANCE ELECTROLYTES FOR LMBs

High concentration electrolytes (HCEs)

The strategy of enhancing electrolyte concentrations to improve battery performance originated from LIB research.[75,76] In 1985, McKinnon and Dahn first discovered that saturated $LiAsF_6$ in propylene carbonate (PC) could prevent the co-intercalation of PC into ZrS_2 electrodes.[77] Jeong et al. also reported in 2003[78] and 2008[79] that concentrated $LiClO_4$, $LiPF_6$, and $LiN(SO_2C_2F_5)_2$ (LiBETI) salts in PC (>2.5 M) could effectively suppress the co-intercalation of PC into graphite anodes. In 2010[80] and 2014,[81] Yamada et al. further

Table 1. Recently reported electrolytes with excellent anode-free battery performances

Electrolyte type	Electrolyte formulation	Electrolyte amount	Cell condition	Cycling condition & capacity retention
HCEs	4 M LiFSI/DME[46]	~44 g/Ah	1.71 mAh/cm^2 Cu‖LFP coin cell	3.8 V, ~C/8 cycling, 60% @ 50 cycles
	7 M LiFSI/FEC[47]	not mentioned	1.83 mAh/cm^2 Cu‖LNMO coin cell	5 V, 54% @ 50 cycles
	3 M LiFSI/DME-DOL[48]	~37 mL/Ah	1.6 mAh/cm^2 Cu‖LFP coin cell	3.8 V, 64% @ 50 cycles
	2 M $LiPF_6$/EC-DEC-FEC[49]	not mentioned	~1.6 mAh/cm^2 Cu‖NMC111 coin cell	4.3 V, ~C/8 cycling, 40% @ 50 cycles
	2 M $LiPF_6$/THF-MTHF[43]	not mentioned	2.3 mAh/cm^2 Cu‖LFP coin cell	3.5 V, ~65% @ 100 cycles
LHCEs	1LiFSI-1.2DME-3TTE[50]	3 g/Ah	~4.2 mAh/cm^2 Cu‖NMC811 coin cell	4.4 V, C/3 cycling, 77% @ 70 cycles
4SEs and derived	1 M LiFSI/FDMB[35]	~2 g/Ah	~200 mAh Cu‖NMC532 pouch cell	4.2 V, 100% DOD, C/5 D/3, 80% @ 100 cycles
	1 M LiFSI/DME-FDMH[51]	~2.1 mL/Ah	~200 mAh Cu‖NMC811 pouch cell	4.2 V, 100% DOD, C/5 D/4, 75% @ 120 cycles
FFEs	1 M $LiPF_6$/FEC-FEMC-D2[52]	~47 g/Ah	~2.0 mAh/cm^2 Cu‖NMC811 coin cell	4.4 V, ~C/4 cycling, 50% @ 30 cycles
	1 M LiFSI/FSA[53]	~25 mL/Ah	1.6 mAh/cm^2 Cu‖NMC622 coin cell	4.3 V, C/10 D/3, 50.8% @45 cycles
Multi-salt electrolytes	1 M LiTFSI + 2 M LiFSI + 3 wt % $LiNO_3$/ DME-DOL[54]	not mentioned	0.85 mAh/cm^2 Cu‖LFP coin cell	3.8 V, C/4 cycling, 39% @ 100 cycles
	1 M LiDFOB + 0.2 M $LiBF_4$/FEC-DEC[55]	~2 g/Ah	~245 mAh Cu‖NMC532 pouch cell	3.6–4.5 V, C/5 D/2, 40°C, 80% @ 90 cycles
	1.8 M LiDFOB + 0.4 M $LiBF_4$/FEC-DEC[56]	~2 g/Ah	~210 mAh Cu‖NMC532 pouch cell	3.6–4.5 V, C/5 D/2, 40°C hot formed, 80% @ 195 cycles (high pressure)
	2 M LiDFOB + 1.4 M $LiBF_4$/FEC-DEC[39]	~2.2 mL/Ah	~230 mAh Cu‖NMC532 pouch cell	3.6–4.5 V, C/5 D/2, 40°C hot formed, 80% @ 200 cycles (high pressure)
Electrolytes with novel additives	1 M $LiPF_6$/FEC-TFEC[57]	3 g/Ah	~230 mAh Cu‖532 pouch cell	3.6–4.5 V, C/5 D/2, 40°C, 80% @ 40 cycles
	1 M $LiPF_6$/EC-DEC with 10 mM $In(OTf)_3$ and 0.5 M $LiNO_3$[58]	8.37g/Ah	4.3 mAh/cm^2 Cu‖NMC811 coin cell	3–4.3 V, 0.1C 0.3D, 50% @ 80 cycles
	1 M LiTFSI + 1.5 M LiFSI /G3-HFE[59]	2.5 g/Ah	~4.85 mAh/cm^2, 2.46 Ah Cu‖NMC811+Li_2O pouch cell	2.8–4.4 V, 100% DOD, 0.375C 0.5D, 80% @ 300 cycles

applied this strategy to other solvents, such as dimethyl sulfoxide (DMSO) and acetonitrile (AN), with 3.2–4 M $LiN(SO_2CF_3)_2$ (LiTFSI) as the salt. By combining density functional theory calculation with molecular dynamics simulations (DFT-MD), they found that HCEs effectively reduced the number of solvent molecules coordinated with Li-ions. This not only contributed to the suppression of co-intercalation but also significantly improved the electrolyte reductive stability. In 2015, Suo et al. applied this strategy to water-based electrolytes, which enabled the first aqueous LIB chemistry.[82]

Jeong et al. first explored the performance of HCEs for Li metal cycling in 2008.[83] They reported an improved CE for Li metal with high concentrations of 3.27 M LiBETI in PC, but the choices of Li salt and solvent were not optimized, and the CE was still below 85%. A significant milestone on HCEs was published in 2015, where Qian et al. reported that 4 M $LiN(SO_2F)_2$ (LiFSI) in 1,2-dimethoxyethane (DME)[34] enabled a high Li cycling CE: 99.1% under 0.2 mA/cm^2 and 98.4% under 4 mA/cm^2. In this work, LiFSI was chosen as the salt due to its high ionic conductivity and solubility, whereas DME was selected as the solvent due to its excellent stability toward LMAs. The high electrolyte concentration significantly reduced the amount of uncoordinated DME and enhanced the coordination of FSI^- anions to Li-ions (Figure 3A). Such solvation environments improved the electrolyte's reductive[34] and oxidative[46,84] stability and allowed them to form inorganic-rich SEIs with lower ionic resistances. The electrolyte not only deposited Li in large granules with excellent cycling efficiencies (Figure 3B) but was also compatible with high-voltage cathodes, such as NMC. The stable cycling of Li with the tuning of the electrolyte formulation has inspired a series of works optimizing HCEs for LMBs.

In 2018, Fan et al. reported HCEs with 10 M LiFSI in dimethyl carbonate (DMC) or ethylene carbonate (EC)-DMC mixture, achieving a high Li cycling CE of 99.3%.[60] Li‖NMC622 full cells were found to stably cycle at 4.6 V with 86% capacity retention after

Table 2. Recently reported electrolytes with good performances for LMBs with excess Li

Electrolyte type	Electrolyte formulation	Electrolyte amount	Cell condition	Cycling condition & capacity retention
HCEs	4 M LiFSI/DME[46]	~37 mL/Ah	Li‖NMC111 (1.5 mAh/cm^2), N/P = 60	4.3 V, C/3 1D, 92% @ 500 cycles
	10 M LiFSI/EC-DMC[60]	not mentioned	Li‖NMC622 (13 mg/cm^2)	4.6 V, 86% @100 cycles
	7 M LiFSI/FEC[47]	not mentioned	Li‖LNMO (1.83 mAh/cm^2), N/P = 1.4	5 V, ~70% @ 140 cycles
	1:2.5 LiFSI/SL[61]	~30 mL/Ah	Li‖LFP (1.76 mAh/cm^2), N/P=2.34	3.8 V, 0.5C cycling, 92.7% @ 140 cycles
	4 M LiFSI/DEE[62]	8 g/Ah	Li‖NMC811 (4.8 mAh/cm^2), N/P = 2	4.4 V, 0.2C 0.3D, 80% @ 182–200 cycles
LHCEs	1.2 M LiFSI/DMC-BTFE[63]	~79 mL/Ah	Li‖NMC111 (2 mAh/cm^2), N/P = 45	4.3 V, 0.5C 2D, 80% @ 700 cycles
	1.2 M LiFSI/TEP-BTFE[64]	~50 mL/Ah	Li‖NMC622 (1.6 mAh/cm^2), N/P = 56.25	4.4 V, C/3 1D, ~97% @ 600 cycles
	1.2 M LiFSI + 0.15 M LiDFOB/EC-EMC-BTFE[65]	~6.6 mL/Ah	Li‖NMC111 (3 mAh/cm^2), N/P = 3.33	4.3 V, C/3 cycling, ~80% @ ~85 cycles
	1LiFSI-3TMS-3TTE[66]	~39 mL/Ah	Li‖NMC (1.5 mAh/cm^2), N/P = 6.67	4.3 V, C/3 cycling, ~80% @ 300 cycles
	1LiFSI-1.2DME-3TTE[50]	3 g/Ah	Li‖NMC811 (4.2 mAh/cm^2), N/P = 2.4	4.4 V, C/3 cycling, 80% @ 155 cycles
	1 M LiFSI/DME/TFEO[42]	50 g/Ah	Li‖NMC811 (1.5 mAh/cm^2), N/P = 6.7	4.4 V, C/3 cycling, 80% @ 300 cycles
4SEs and derived	1 M LiFSI/FDMB[35]	~30 g/Ah	Li‖NMC532 (1.6 mAh/cm^2), N/P = 6	4.2 V, C/3 cycling, 90% @ 400 cycles
	1 M LiFSI/FDMB[35]	~6 g/Ah	Li‖NMC532 (1.6 mAh/cm^2), N/P = 2.5	4.2 V, C/3 cycling, 100% @ 210 cycles
	1 M LiFSI/DME-FDMH[51]	~8 mL/Ah	Li‖NMC532 (2.5 mAh/cm^2), N/P = 1.6	4.2 V, C/3 cycling, 84% @ 250 cycles
	1 M LiFSI/DME-FDMH[51]	~10 mL/Ah	Li‖NMC811 (2 mAh/cm^2), N/P = 2	4.4 V, C/2 cycling, 76% @ 250 cycles
FFEs	1 M $LiPF_6$/FEC-FEMC-D2[52]	~50 g/Ah	Li‖NMC811 (2 mAh/cm^2), N/P = 1	4.4 V, ~C/2 cycling, 95% @ 120 cycles
	1 M LiFSI/FSA[53]	~25 mL/Ah	Li‖NMC622 (1.6 mAh/cm^2), N/P = 7.6	4.3 V, 89% @ 200 cycles
	1 M LiFSI/DMTMSA[67]	2.62 g/Ah	Li‖NMC811 (4.86 mAh/cm^2), N/P = 0.39	4.7 V, 0.15C 0.5D, 88% @ 90 cycles
Multi-salt electrolytes	2 M LiTFSI + 2 M LiDFOB/DME[68]	~40 mL/Ah	Li‖NMC111 (1.7 mAh/cm^2), N/P = 30	4.3 V, C/3 1D, 79% @ 500 cycles
	1 M LiTFSI + 2 M LiFSI + 3 wt % $LiNO_3$/DME-DOL[54]	Not mentioned	Li‖LFP (0.85 mAh/cm^2), N/P = 0.44	3.8 V, C/4 cycling, 83% @ 100 cycles
Electrolytes with novel additives	0.5 M $LiPF_6$/EC-DEC with $LiNO_3$[69]	~80 mL/Ah	Li‖NMC (8 mg/cm^2), N/P = 8.4	4.3 V, ~1C cycling, ~80% @ 250 cycles
	1.0 M $LiPF_6$/FEC-DMC-DME with 1.1 wt % $LiNO_3$[70]	Not mentioned	Li‖LFP (2 mAh/cm^2), N/P = 5	4 V, 0.2C cycling, ~85% @ 120 cycles
	2.3 M LiFSI/DME with 20 mM of $LiNO_3$ and 20 mM of CuF_2[71]	4 mL/Ah	Li‖LFP (2 mAh/cm^2), N/P = 5	4 V, 0.2C cycling, 100% @ 80 cycles
	1 M $LiPF_6$/EC-DEC with 10 mM $In(OTf)_3$ and 0.5 M $LiNO_3$[58]	8.37 g/Ah	Li‖NMC811 (4.3 mAh/cm^2), N/P = 2.09	4.3 V, 0.3C 0.5D, 80% @ 160 cycles
	1 M $LiPF_6$/FEC-EMC with 3 wt % $LiNO_3$ and 1 wt % TPFPB[72]	3.4 g/Ah	Li‖NMC811 (3.4 mAh/cm^2), N/P = 2.68	4.4 V, 0.1C 0.3D, 80% @140 cycles
	0.8 M $LiPF_6$/FEC-DMC with 5 wt % 4 M $LiNO_3$/DMSO[73]	~7.9 mL/Ah	Li‖NMC811 (2.5 mAh/cm^2), N/P = 4	4.4 V, 0.5C cycling, 75% @ 200 cycles
	0.8 M LiTFSI + 0.2 M LiDFOB + 0.05 M $LiPF_6$/EMC-FEC (v/v, 3:1) with 1 wt % P_2S_5-saturated CS_2[74]	4 mL/Ah	Li‖NCMA (4 mAh/cm^2), N/P = 5	4.3 V, 0.2C 0.5D, ~80% @ 130 cycles

100 cycles. Suo et al. reported 7 M LiFSI in fluoroethylene carbonate (FEC) electrolyte had an excellent CE of 99.6%,[47] although it needs ~300 cycles for the CE to gradually increase and stabilize. Li‖LNMO cells with limited excess Li cycled at 5 V with ~70% capacity retention after 140 cycles. Zeng et al. reported that LiFSI in triethyl phosphate (TEP) with a molar ratio of 1:2 had a CE for Li metal over 99%,[85] but the performance of Li metal full cells was not demonstrated. In 2019, Maeyoshi et al. reported 1:2.5 (molar ratio) LiFSI in tetramethylene sulfone (TMS, also known as sulfolane, SL) had a CE of 98%. Li‖LFP cells lasted 140 cycles with 92.7% capacity retention.[61] Besides HCEs, there were also several reports using moderately concentrated (2–3 M) electrolyte formulations. In 2019, Hwang and co-workers reported 3 M LiFSI/DME-DOL[48] and 2 M $LiPF_6$/EC-DEC-FEC[49] with enhanced anode-free battery performance. In 2020, Wang and co-workers[43] proposed a 2 M $LiPF_6$ in tetrahydrofuran (THF)-2-methyl tetrahydrofuran (MTHF) mixture, achieving record high Li metal CE of 99.83% on a bismuth (Bi)-graphite (Gr) composite substrate. This high CE was attributed to a LiF/organic bilayer SEI with a

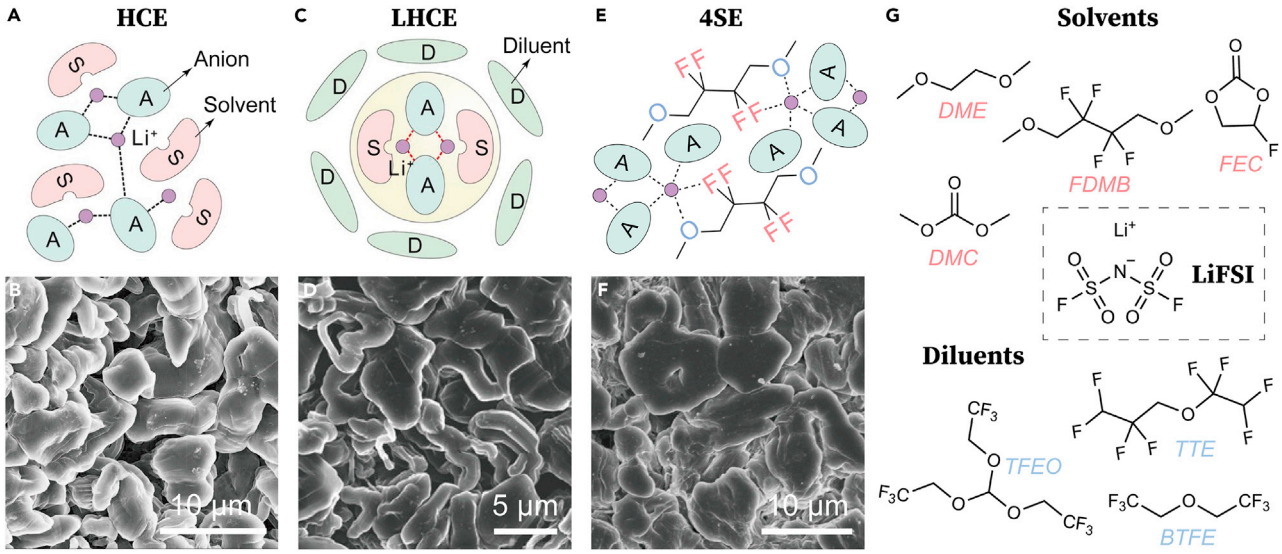


Figure 3. HCEs, LHCEs, and 4SEs

(A) Schematic of the solvation environment in an HCE.

(B) Li metal deposition morphology in 4 M LiFSI/DME. Reproduced with permission.[34] Copyright 2015, Nature Publishing Group.

(C) Schematic of the solvation environment in an LHCE.

(D) Li deposition morphology in 1.2 M LiFSI/DMC-BTFE. Reproduced with permission.[63] Copyright 2018, Wiley.

(E) Schematic of the solvation environment in 1 M LiFSI/FDMB.

(F) Li deposition morphology in 1 M LiFSI/FDMB. Reproduced with permission.[35] Copyright 2020, Nature Publishing Group.

(G) Solvents, diluents, and Li salt (LiFSI) commonly used in these electrolytes.

nearly pure LiF inner layer directly contacting metallic Li to promote uniform deposition. The Bi-Gr||LFP cell maintained ~65% capacity after 100 cycles, indicating a full-cell CE of ~99.6%.

HCEs brought LMB a step closer to practical applications. HCEs for LMBs with limited or zero (anode-free) excess Li started to become cyclable. However, HCEs have several drawbacks. First, HCEs usually have reduced ionic conductivity compared with standard-concentration (1M) electrolytes. For example, by increasing the concentration of LiFSI in DME from 1 to 4 M, the ionic conductivity is reduced by a factor of three.[34] Second, electrolyte viscosity inevitably increases under high concentrations. Therefore, HCEs require specially designed separators[75] and sometimes have poor wettability to the porous cathodes.[6] These drawbacks could possibly lead to reduced full-cell rate capability.[47] Furthermore, HCEs can increase the electrolyte cost by more than 400%[6] due to the high content of expensive Li salts.

Localized high concentration electrolytes (LHCEs)

To address the drawbacks of HCEs, the most straightforward approach is to dilute HCEs to standard concentrations with another solvent. Although it is nontrivial to choose the added solvent since it should not disrupt the Li-ion solvation environments in HCEs, several recent works have identified a few candidates. In 2013, Dokko et al. reported that high concentration LiTFSI in triglyme (G3)/tetraglyme (G4) diluted with a hydrofluoroether 1,1,2,2-tetrafluoroethyl-2,2,3,3-tetrafluoropropyl ether (TTE) improved the cyclability of Li-sulfur (S) batteries due to the suppression of Li polysulfide dissolution.[86] The addition of TTE effectively reduced the electrolyte viscosity and enhanced its ionic conductivity without disrupting the solvation structure between glyme and LiTFSI. In 2015, Watanabe and co-workers reported a similar formulation with an optimized solvent ratio giving good performance of both LIBs and Li-S batteries.[87] In 2016, they

systematically compared a series of solvents to dilute HCEs, with the selection of toluene and TTE as the best candidates.[88] In 2017, Doi et al. reported a highly concentrated $LiBF_4$ in PC diluted by TTE, enabling high reversibility of LNMO cathodes.[89] However, none of these works explicitly demonstrated the impact of diluted HCEs on LMA cycling.

In 2018, based on their previous works on HCEs, Chen and co-workers reported a Li metal CE of 99.3% in 5.5 M LiFSI/DMC diluted with bis(2,2,2-trifluoroethyl) ether (BTFE).[63] Through MD simulation and Raman spectroscopy, they showed that BTFE did not participate in the direct solvation of Li-ions, whose local environment, thus, resembled those in the HCE case (Figure 3C). Therefore, they named this electrolyte a LHCE and BTFE a diluent (Figure 3G). This LHCE electrolyte enabled uniform Li deposition morphology (Figure 3D) and excellent stability toward high-voltage cathodes with a much lower equivalent concentration (1.2 M) compared with HCEs to reduce viscosity and cost. In particular, a Li||NMC111 full cell maintains 80% capacity retention after 700 cycles. This work suggests that the diluents need to be highly stable, highly miscible with co-solvents, and weakly interacting with Li-ions. Following this work, a series of LHCEs were reported, including 1.2 M LiFSI in TEP-BTFE with good fire retardency,[64] 1.28 M LiFSI in FEC-methyl (2,2,2-trifluoroethyl) carbonate (FEMC)-tetrafluoro-1-(2,2,2-trifluoroethoxy)ethane (D2) and 0.7 M LiBETI in FEC-diethyl carbonate (DEC)-methoxyperfluorobutane (M3) with good extreme temperature performance,[90] 1.2 M LiFSI with 0.15 M $LiBF_2(C_2O_4)$ (LiDFOB) in EC-ethyl methyl carbonate (EMC)-BTFE,[65] LiFSI in TMS-TTE,[66] LiFSI in DME-TTE,[50] and 1M LiFSI in DME/tris(2,2,2-trifluoroethyl)orthoformate (TFEO).[42] Among them, the 1LiFSI-1.2DME-3TTE (molar ratio)[50] achieved the best LMB cycling performance, with 99.3% CE for 300 cycles and uniform Li deposition. A Li||NMC811 (4.2 mAh/cm^2) coin cell with limited excess Li (N/P ratio = 2.4) was able to maintain 80% capacity after 155 cycles under lean electrolyte condition (3 g/Ah). Even when the excess Li was replaced by bare Cu foil, 77% of capacity was still retained after 70 cycles. This was a real advance in electrolyte design because LMBs became cyclable under realistic conditions that include high cathode loadings, limited or zero excess Li, and lean electrolyte amounts.[44] Although the cycle life still fell short, the LHCE strategy proved to be beneficial for LMBs, and further optimizations of salt and solvents could bring further gains in performance.

Single-salt-single-solvent electrolytes (4SEs)

In LHCEs, one good solvent is used to solvate Li-ions, whereas the other poor solvent dilutes the concentration without compromising electrolyte stability. To simplify the electrolyte composition and conduct an in-depth study on structure-performance relationships, Yu and co-workers designed a new single-salt-single-solvent, standard-concentration electrolyte (4SE), which was enabled by a rationally designed solvent molecule that could simultaneously possess high stability and Li-ion solvability.[35] The new solvent is a fluorinated analog of DME, where the alkyl chain was first prolonged and then decorated with –F groups, resulting in the fluorinated 1,4-dimethoxylbutane (FDMB). The –F groups were selectively functionalized on the two carbon atoms at the center, maintaining their distances to –O– groups so that FDMB could dissolve Li salts. As a result, Li-ions coordinated with O and F simultaneously to form a five-atom ring structure (Figure 3E), which gave the electrolyte a unique brownish color. With LiFSI as the salt, the 1 M LiFSI/FDMB electrolyte achieved homogeneous Li deposition (Figure 3F) and an excellent CE of 99.5%. In Cu||Li half cells, the CE reached 99% within 5 cycles, in contrast to tens or hundreds of cycles for HCEs and LHCEs. This was made possible by eliminating any unstable solvents, such as DME and DMC, in the electrolyte. Meanwhile, 1 M LiFSI/FDMB showed a high oxidative stability of ~6 V. As a result, a Li||NMC532 cell (N/P =

2.5) maintained 100% capacity after 200 cycles. A Cu||NMC532 industrial anode-free pouch cell maintained 80% capacity after 100 cycles. Later, shifting to the more stable fluorinated 1,6-dimethoxyhexane (FDMH) system,[51] an even longer cycle life was achieved with improved kinetics. The solvents in 4SEs can be not only the partially fluorinated ones, such as FDMB and FDMH, but also fluorine-free weakly solvating ones, such as 1,2-diethoxyethane (DEE) by Chen et al.[62] and 1,4-dioxane (1,4-DX), MTHF, and 2,5-dimethyl-tetrahydrofuran (DMeTHF) by Xu et al.[91] These works demonstrated the feasibility of molecular design to manipulate solvent physiochemical properties. Following this strategy, new solvent molecules are expected to be developed to further improve LMB performances.

Fully fluorinated electrolytes (FFEs)

Increasing fluorination degree is considered generally beneficial for LMBs due to their high oxidative stability and F-rich SEI formation.[52,53] Therefore, the heavily or even fully fluorinated electrolyte chemistries were studied. In 2018, Fan et al. reported 1 M $LiPF_6$ in FEC-FEMC-D2 (2:6:2 by weight)[52] with Li metal CE of 99.2%. With the fully fluorinated electrolyte components, the highly fluorinated interphases are formed on both Li metal and high-voltage cathodes, with substantial amount of LiF. An anion-rich Li-ion environment induced by the weak solvation of highly fluorinated solvents was critical in enabling this interphase chemistry. As a result, a Li||NMC811 (2 mAh/cm^2) cell with one-fold excess Li maintained 95% capacity after 120 cycles. Furthermore, the –F groups significantly improved the electrolyte non-flammability by inhibiting the oxygen radical propagation during combustion. In 2020, Xue et al. reported 1 M LiFSI in *N*,*N*-dimethylsulfamoyl fluoride (FSA)[53] with Li metal CE of 99.03% and fast activation within 10 cycles. This special solvent not only was fluorinated but also possessed the fluorosulfonyl group ($–SO_2F$) resembling the FSI^- anion. It was named a "full fluorosulfonyl" (FFS) electrolyte (Figure 4A). With this design, SEI primarily originated from the $–SO_2F$ decomposition, with a significant amount of LiF. This enabled uniform Li deposition morphology (Figure 4B) and improved CE. As a result, a Li||NMC622 (1.6 mAh/cm^2) cell maintained 89% capacity after 200 cycles. In 2021, Xue et al. further proposed 1 M LiFSI in *N*,*N*-dimethyltrifluoromethane-sulfonamide (DMTMSA),[67] which replaced the –F group in the previous FSA solvent by a $–CF_3$ group. This design further promoted the oxidative stability of the electrolyte. A Li||NMC811 (4.86 mAh/cm^2, N/P ratio = 0.39) cell cycled at 4.7 V maintained 88% of capacity after 90 cycles.

Multi-salt electrolytes

Besides solvent, Li salts are another component of the electrolyte that can be tuned to improve the cyclability of LMBs. Promising work has combined different functionalities of two or three Li salts to simultaneously address requirements, such as Li metal compatibility and high voltage stability. In most cases, each salt in the multi-salt system possesses its unique contribution, and they function synergistically. Therefore, multi-salt systems provided more tunabilities and functionalities compared with 4SEs. In 2018, Jiao et al. reported that 2 M LiDFOB with 2 M LiTFSI in DME enabled a Li||NMC811 (1.7 mAh/cm^2) cell to maintain 79% capacity after 500 cycles.[68] In this design, the electrolyte was superconcentrated so that the Li salts predominantly determines the interphase chemistries. LiDFOB was selected for its effect in passivating high-voltage cathodes with an insoluble, boron-rich, polymeric cathode-electrolyte interface (CEI) layer,[92] whereas LiTFSI was a known beneficial Li salt that promoted the formation of polymeric species $–(CH_2–CH_2–O)_n–$ in the SEI to protect LMAs. Qui et al. Reported a ternary-salt electrolyte using 1 M LiTFSI, 2 M LiFSI, and 3 wt % $LiNO_3$ in DME-DOL (1:1 volume ratio), enabling 99.1% CE.[54] In this electrolyte, $LiNO_3$ and LiFSI promoted the formation of Li_2O[69,93] and LiF[94] species in the SEI respectively, whereas more stable LiTFSI salt slowed down DOL polymerization at high concentrations.

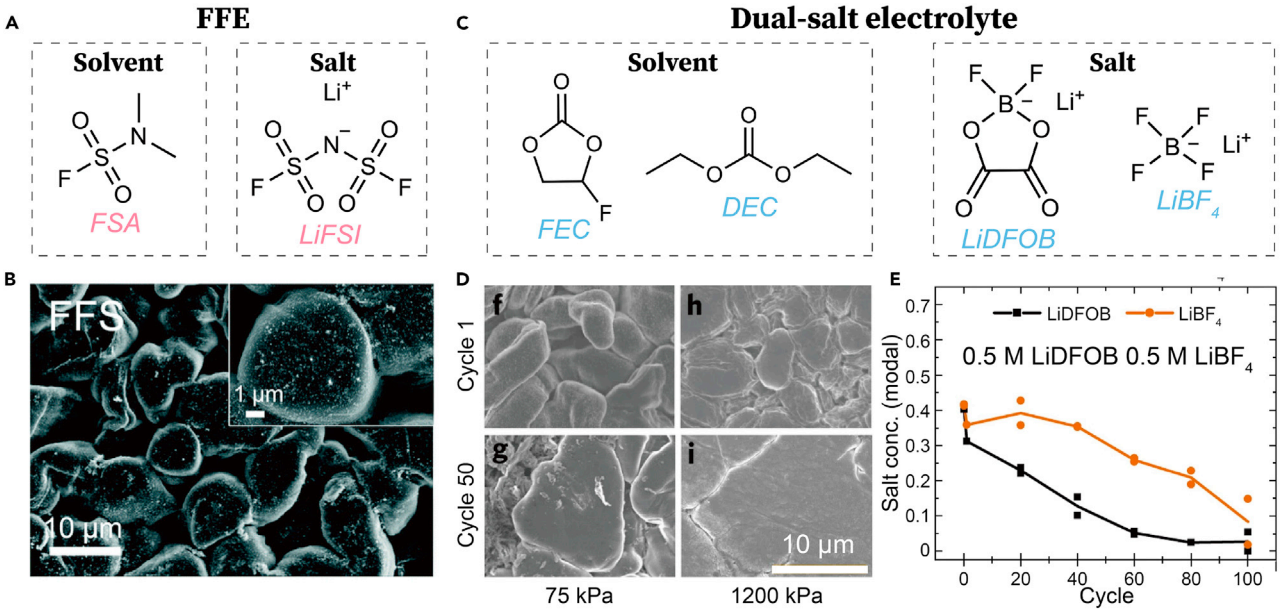


Figure 4. FFEs and multi-salt electrolytes

(A) Chemical composition of a "full fluorosulfonyl" electrolyte (FFS).

(B) Li deposition morphology in FFS. (A and B) Reproduced with permission.[53] Copyright 2020, Royal Society of Chemistry.

(C) Chemical composition of a dual-salt electrolyte.

(D) Li deposition morphology in the dual-salt electrolyte under different pressures.

(E) Salt concentration evolution results after long-term cycling. (D and E) Reproduced with permission.[55] Copyright 2019, Nature Publishing Group.

In 2019, Dahn and co-workers[55] reported 1 M LiDFOB and 0.2 M $LiBF_4$ in FEC-DEC (1:2 by volume ratio, Figures 4C and 4D), enabling an anode-free Cu||NMC532 pouch cell to retain 80% capacity after 90 cycles under a lean electrolyte condition. The authors chose LiDFOB because it promoted the LMA cycling performances, especially when combined with FEC.[95–97] The addition of $LiBF_4$ further improved the cell capacity retention by alleviating the fast decomposition and consumption of LiDFOB during cycling. Interphase characterization showed an increased amount of organic F species in the SEI together with inorganic LiF components. Nuclear magnetic resonance (NMR) measurement showed that both LiDFOB and $LiBF_4$ concentrations continuously decreased during cycling until full consumption (Figure 4E), whereas the solvents were mostly preserved. The authors, thus, attributed the cell failure to the loss of Li salt for ideal SEI formation and Li-ion transfer. Therefore, in follow-up works,[39,56] the authors increased the salt concentrations and achieved even longer cycle life. With 1.8 M LiDFOB and 0.4 M $LiBF_4$,[56] a Cu||NMC532 pouch cell hot formed at 40°C lasted 195 cycles with 80% capacity retention. With even higher concentration, 2 M LiDFOB and 1.4 M $LiBF_4$,[39] 200 cycles with 80% capacity retention were achieved, which represents one of the best anode-free battery cycling performances. It is worth noting that these cells were cycled between 3.6–4.5 V, so that the depth of discharge (DOD) was ~70%–80%. This enabled a Li reservoir on the Cu surface after the first cycle to compensate subsequent Li loss but would moderately undermine the battery energy.

Electrolytes with novel additives

Electrolyte additives have long been proven to impact the performance of Li based batteries.[98–100] In the 1980–1990s, there were reports using 2-methylfuran[101] or CO_2[102,103] to improve the CE of Li. Various inorganic metal ions such as Zn^{2+}, In^{3+}, and Ga^{3+} that could deposit and then alloy with metallic Li were also used as

additives.[104] After 2000, other metal ions, such as Na^+, K^+, and Mg^{2+} were also explored as additives to be reduced and deposited at a higher potential than Li[105–107] and form protective coatings. In 2013, Ding et al. reported[108] that the positively charged Cs^+ can electrostatically shield the Li metal surface, resulting in uniform Li deposition. In 2015, Qian et al.[109] used water as the electrolyte additive for carbonate-based electrolytes, which led to compact nanorod-structured deposited Li. Nevertheless, even with these additives, Li metal CE rarely exceeded 95%.

During the last 5 years, various types of new organic/inorganic additives and their combinations have been explored and optimized. These additives include solvents (vinyl carbonate [VC],[110,111] FEC,[70,111,112] bis(2,2,2-trifluoroethyl) carbonate (TFEC),[57] etc.), salts ($LiAsF_6$,[110] $LiPF_6$,[111] $LiNO_3$,[36,70,71,93,113–115] CuF_2,[71] indium trifluoromethanesulfonate ($In(OTf)_3$),[58] etc.), and other materials, including LiF,[116] nanodiamonds,[117] Li polysulfides (LiPS),[36] and tris(penta-fluorophenyl)borane (TPFPB).[72] Among them, fluorinated chemicals, such as LiF, FEC, and CuF_2, were particularly effective. Various characterization methods showed that these additives induce either LiF-rich[71,112] or highly homogeneous multilayered SEIs.[118,119] Meanwhile, $LiNO_3$ has been broadly adopted in various works as one of the most effective additives. NO_3^- has a high cathodic decomposition potential of ~1.7 V versus Li/Li^+, and its decomposition led to a nitrogen/Li_2O rich multilayered SEI and promoted spherical nuclei formation and homogeneous growth of Li metal with a high CE of 98%.[69,113] In 2018, Liu et al. reported improved CE and a Li||NMC111 (8 mg/cm^2) cell with over 250 cycles in a carbonate electrolyte with sustained release of $LiNO_3$ from a polymeric membrane[69] (Figures 5A–5C). Combining $LiNO_3$ with LiPS results in an even higher CE of 99.1%, which was attributed to the formation of Li_2S/Li_2S_2 in the outer layer of the SEI.[36] However, no full-cell performance was reported due to the poor oxidative stability of the ether-based electrolyte used in this work. Combining $LiNO_3$ with FEC in a carbonate-ether mixed electrolyte, Zhang et al.[70] reported a Li||LFP (2 mAh/cm^2) pouch cell that maintained ~85% capacity after 120 cycles. In 2019, combining $LiNO_3$ with CuF_2 in moderately concentrated 2.3 M LiFSI/DME, Yan et al.[71] achieved 99.5% CE and highly cyclable Li||LFP pouch cell with 4 mL/Ah of electrolyte and attributed the improved performance to the preferential absorption of Cu^{2+}-NO_3^- complexes within the inner Helmholtz plane that regulated the SEI formation.

In 2020, based on the CuF_2 dissolution promoter,[93] Lu and co-workers[58,72] explored the synergetic effect of $LiNO_3$ with Lewis-acid based additives that could coordinate with NO_3^- and help dissociate $LiNO_3$ within carbonate-based electrolytes, where $LiNO_3$ could barely dissolve by themselves. They combined $LiNO_3$ with $In(OTf)_3$ in 1 M $LiPF_6$/EC-DEC, finding a wavy-nanostructured SEI with high content of inorganic nanocrystallites (Li_3N and Li_2O).[58] A Li||NMC811 (4.3 mAh/cm^2) coin cell (N/P = 2.09) maintained 80% capacity after 160 cycles with 8.37 g/Ah of electrolyte. In another work,[72] they combined $LiNO_3$ with TPFPB (Figures 5D and 5E) in 1 M $LiPF_6$/FEC-EMC, enabling a Li||NMC811 (3.4 mAh/cm^2) cell (N/P = 2.68) to maintain 80% capacity after 140 cycles with 3.4 g/Ah of electrolyte. TPFPB could absorb NO_3^- to reduce its abundance in the first solvation shell of Li-ions (Figure 5F). TPFPB could also decompose at the cathode surface to form F and B rich interphases, enabling the cyclability of high-voltage cathodes including LNMO. Beside these works, Wang and co-workers[73] reported a 0.8 M $LiPF_6$/FEC-DMC electrolyte with 5 wt % of 4 M $LiNO_3$/DMSO. Due to the high donor number of DMSO, high concentration of $LiNO_3$ could be dissociated and incorporated into the carbonate-based electrolyte without precipitation. As a result, an inorganic-rich SEI was observed with 99.55% Li metal CE. In 2021, Sun and co-workers[74] reported 1 wt % of P_2S_5-saturated CS_2 (PSC)

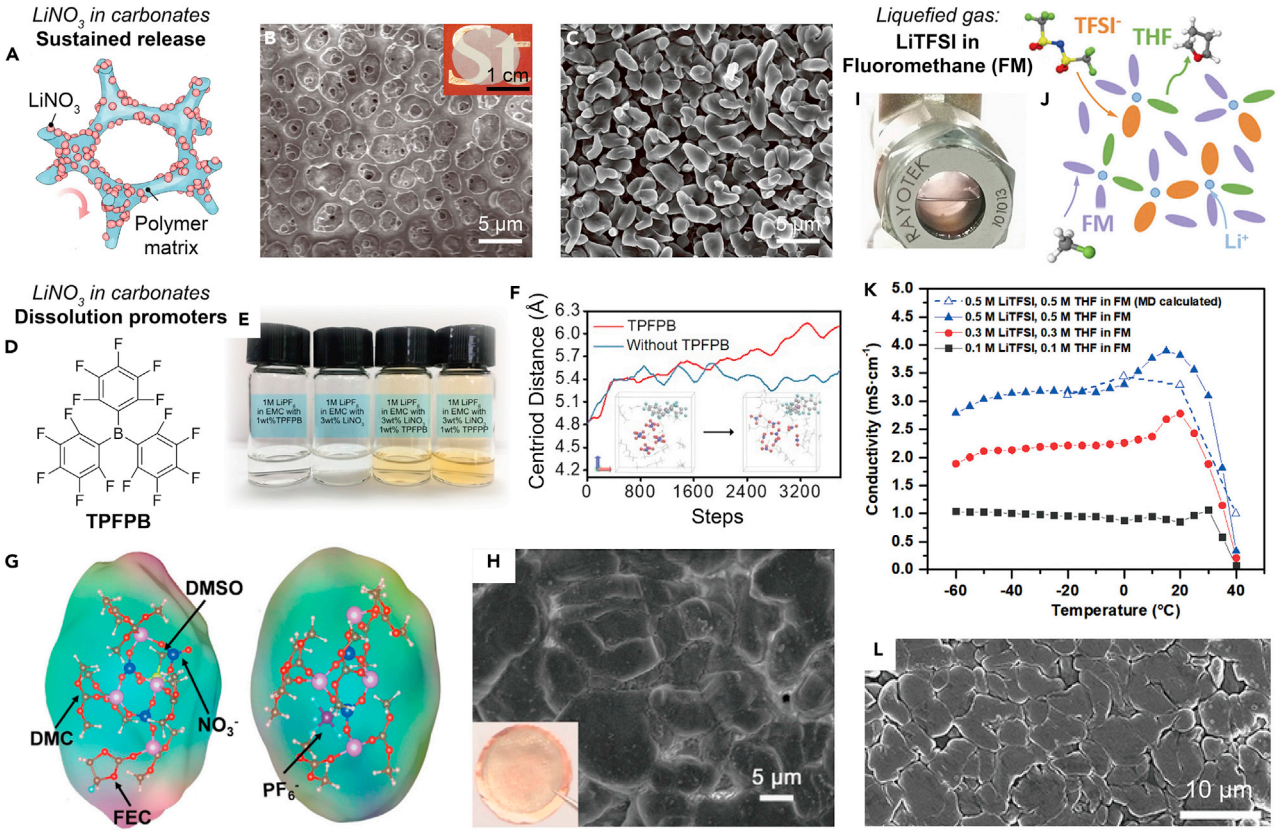


Figure 5. Electrolytes with $LiNO_3$ additive and liquefied gas electrolytes

(A) Schematic of a polymeric membrane that enables sustained $LiNO_3$ release into the electrolyte.

(B) SEM image of the membrane.

(C) Li deposition morphology in 0.5 M $LiPF_6$/EC-DEC with $LiNO_3$. (A–C) Reproduced with permission.[69] Copyright 2018, Nature Publishing Group.

(D) A TPFPB molecule that promotes $LiNO_3$ dissolution in carbonate electrolytes.

(E) Digital photo of electrolytes w/o dissolution promoters.

(F) Simulated solvation environment in a TPFPB/$LiNO_3$ containing electrolyte. (D–F) Reproduced with permission.[72] Copyright 2020, Wiley.

(G and H) Solvation structure and Li deposition morphology in 0.8 M $LiPF_6$/FEC-DMC with 5 wt % of 4 M $LiNO_3$/DMSO. (G and H) Reproduced with permission.[73] Copyright 2021, Wiley.

(I) Digital photo of a liquefied gas electrolyte 0.3 M LiTFSI-0.3 M THF/fluoromethane (FM).

(J) Solvation environment of the liquefied gas electrolyte.

(K) Ionic conductivity of liquefied gas electrolytes.

(L) Li deposition morphology in 0.3 M LiTFSI-0.3 M THF/FM-CO_2 (19:1). (I–L) Reproduced with permission.[120] Copyright 2019, Elsevier.

as an additive in a carbonate-based electrolyte enabled highly ionically conductive SEI with inorganic L-P-S compounds. With this design, uniform Li deposition was realized and a Li||NCMA (4 mAh/cm^2) cell (N/P ratio = 5) maintained ~80% capacity after 130 cycles.

Overall, electrolyte additives have been shown to be effective, but they are likely to be gradually consumed due to repetitive SEI formation. Nevertheless, the abundance in the additive candidates and their numerous combinations and synergies still pose great potential in improving the performance of LMBs.

Liquefied gas electrolytes

In 2017, Meng and co-workers[121] reported the use of gas-phase materials (under room temperature [RT]) as the electrolyte solvent for LMBs. They found the moderately polar molecule fluoromethane (CH_3F) is gaseous at RT (boiling point: −78°C)

but could liquefy and solvate LiTFSI under controlled temperature and pressure. An electrolyte composed of 0.1 M LiTFSI/CH_3F showed a Li-ion conductivity of around 1 mS/cm within a broad temperature range from −60°C to 30°C. This made the electrolyte an excellent candidate for low-temperature batteries. Meanwhile, when the temperature surpassed ~40°C, salt precipitated to trigger a severe drop in ionic conductivity, which could be utilized to shut down batteries during thermal runaways. With a further optimized electrolyte formulation 0.2 M LiTFSI/CH_3F-CO_2 (19:1), an LCO cathode could normally cycle under −40°C. Furthermore, Li metal showed a CE of 97% within this electrolyte, which was attributed to the LiF and $LiCO_3$ rich SEI from solvent decomposition. In a follow-up work,[120] the LiTFSI solubility was improved by adding THF as a co-solvent. With an optimized formulation 0.3 M LiTFSI-0.3 M THF/CH_3F-CO_2 (19:1, Figures 5I and 5J), the ionic conductivity was further increased to ~2 mS/cm (Figure 5K) with excellent performance at low temperature. This modified electrolyte also promoted a Li metal CE of 99.6% because of the highly uniform and dense Li deposition pattern (Figures 5L and 5M). A $Li_4Ti_5O_{12}$-$LiNi_{0.8}Co_{0.15}Al_{0.05}O_2$ (LTO/NCA) 18650-type full cell was demonstrated for 35 cycles using this electrolyte. In another follow-up work,[122] liquefied LiTFSI/AN-CH_3F was also demonstrated to be effective in LMBs. Overall, liquefied gas electrolytes could significantly promote the low-temperature performances and safety of both LIBs and LMBs, but their long-term cycling stability requires further verifications and the challenges they pose to battery manufacturing needs to be fully addressed for practical applications.

KEY FEATURES ENABLED BY ADVANCED LMB ELECTROLYTES

To achieve a practical LMB, the electrolyte must be compatible with both LMAs and high-voltage cathodes. On the anode side, to achieve an Li metal cycling CE over 99.5%, two sources of Li loss need to be suppressed: "dead Li" and SEI formation.[37] On the cathode side, the electrolyte must possess reasonable oxidative stability. In this section, we try to summarize the key features of recent high-performing electrolytes to meet the requirements above.

Fluorine chemistry

Fluorinated molecules have become increasingly critical in advanced LMB electrolytes. Fluorine is the most electronegative element; therefore, functionalization of solvent molecules with strongly electron-withdrawing –F groups can significantly enhance their oxidative stabilities. This is particularly important for ether-based solvents that are beneficial for LMAs but generally have poor stability toward cathodes.[32] In the past 5 years, a series of fluorinated ether molecules were explored to enable advanced LMB electrolytes, especially for the LHCE strategy. The electron-withdrawing nature of –F group typically weakens the polarity of the solvating groups that interact with Li^+, enabling effectively non-solvating diluents. These diluent molecules included BTFE,[63] TTE,[50] TFEO,[42] D2, M3,[90] etc. However, the choices were limited by commercial availability so that other properties, such as Li-ion solvability, could not be finely tuned at will. In 2020, Yu and co-workers[35] designed a new molecule (FDMB) and its analogs. The rationally designed solvent molecules enabled excellent oxidative stability and tunable Li-ion solvability through selected-site fluorination. This work shows the methodology of designing fluorinated solvent molecules for practical LMBs.

Non-dendritic Li deposition morphology

Homogeneous Li deposition morphology with large Li granular sizes is a necessity for high Li metal CE. Large Li granules with low aspect ratio and tortuosity minimize "dead Li" formation during Li stripping.[37,38] Furthermore, larger grain sizes lead to

reduced surface area, which could minimize SEI formation. Within recent advanced LMB electrolytes, non-dendritic, chunky Li deposition was commonly observed, although with slightly different morphological features. In HCEs, such as 4 M LiFSI/DME,[34] the deposited Li was described as "nodule-like particles with round-shaped edges and ~10 μm sizes" (Figure 3C). In LHCEs, such as 1LiFSI-1.2DME-3TTE, "quite uniform large Li deposits formed"[50] and their morphology seemed slightly chunkier with even lower porosity than that in the HCE (Figure 3E). Various electrolytes with $LiNO_3$ additive were found to enable spherical-shaped Li nuclei[123] that "evolve into nodule-like shapes under increased capacities."[69] In the recently reported 1 M LiFSI/FDMB, Li showed "densely packed, flat, large grains"[35] (Figure 3H). In the dual-salt LiDFOB/$LiBF_4$ electrolyte systems, Dahn and co-workers reported a "dendrite-free, close packed"[55] Li morphology (Figure 4E). With a high operational pressure of 1,200 kPa, this electrolyte resulted in a "smooth Li mosaic comprised of densely packed domains up to 50 μm."[55] The fundamental origin of these similarly uniform Li depositions within different advanced electrolytes is still not fully understood, but they were generally attributed to the altered SEI chemical/structural uniformity and growth rate.

SEI microstructure and homogeneity

With the development of cryogenic electron microscopy (Cryo-EM) technology, highly sensitive materials, such as metallic Li and SEIs, were characterizable at high resolutions.[119,124,125] Li et al.[119] reported the evolution from a mosaic[126] to a multi-layered[127] SEI microstructure after the addition of FEC to 1 M $LiPF_6$/EC-DEC electrolyte (Figures 6A and 6B). In a follow-up work,[118] mosaic SEIs were shown to have high crystallinity "hot-spots" with higher ionic conductivity, which led to aggravated "dead Li" formation during stripping. In contrast, the improved structural homogeneity of multilayered SEIs improved the CE. Cryo-EM was then applied to examine the SEI for several advanced LMB electrolytes. In a LHCE electrolyte (1 M LiFSI/DME-TFEO), Cao et al.[42] described the SEI as "monolithic," because it possessed no crystalline domains and had a consistent thickness of ~10 nm (Figures 6C and 6D). Together with XPS results, this SEI was shown to be uniform in both its thickness and its chemical composition. A similar SEI was observed for 1 M LiFSI/FDMB[35] where the SEI thickness was ~6 nm (Figure 6F), much thinner than that in 1 M LiFSI/DME (~10 nm)[35] (Figure 6E) and carbonate electrolytes (15–20 nm).[119] Furthermore, the SEI was highly uniform, in contrast to that in 1 M LiFSI/DME containing wrinkles. In summary, advanced electrolytes generally enable thin, fully amorphous, and highly homogeneous SEIs.

SEI chemistry and Li-ion solvation environments

The chemical components of SEIs determine critical properties such as ionic conductivity[118] and interfacial energy,[128] thus, impacting Li deposition morphology and CE. The importance of SEI chemistry can be traced back to the Li-ion solvation structure, because the composition of the first solvation shells directly determines the chemical content of the SEI. In this section, we categorize and discuss the common SEI chemistries observed in advanced LMB electrolytes and their origins from the perspective of Li-ion solvation environments.

Inorganic-rich SEIs

SEIs formed in baseline carbonate electrolytes typically exhibit highly organic nature[42,50,52] with ~40% atomic ratio of carbon on the surface of deposited Li (Figure 7A). This is because carbonate solvents coordinate strongly with Li-ions,[129,130] and they significantly decompose during Li deposition. By replacing carbonates with weaker solvating ethers,[42] such as DME, the carbon ratio within SEIs slightly decreased to ~20% (Figure 7B) due to less solvent decomposition.

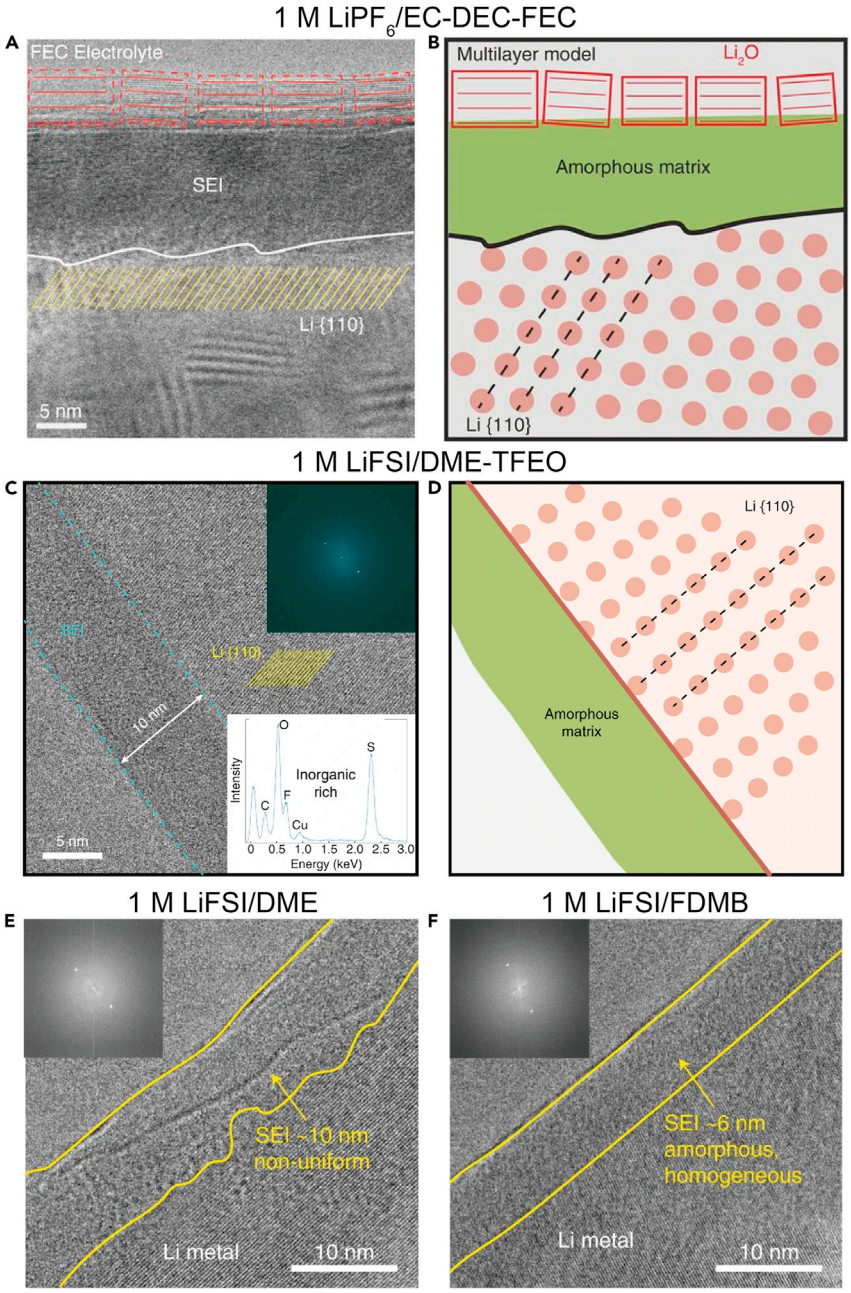


Figure 6. Cryo-EM characterized SEI microstructures

(A) Image of the Li SEI in 1 M $LiPF_6$/EC-DEC-FEC.

(B) Schematic illustration of (A). (A and B) Reproduced with permission.[119] Copyright 2017, American Association for the Advancement of Science.

(C) Image of the Li SEI in 1 M LiFSI/DME-TFEO.

(D) Schematic illustration of (C). (C and D) Reproduced with permission.[42] Copyright 2019, Nature Publishing Group.

(E) Image of the Li SEI in 1 M LiFSI/DME.

(F) Image of the Li SEI in 1 M LiFSI/FDMB. (E and F) Reproduced with permission.[35] Copyright 2020, Nature Publishing Group.

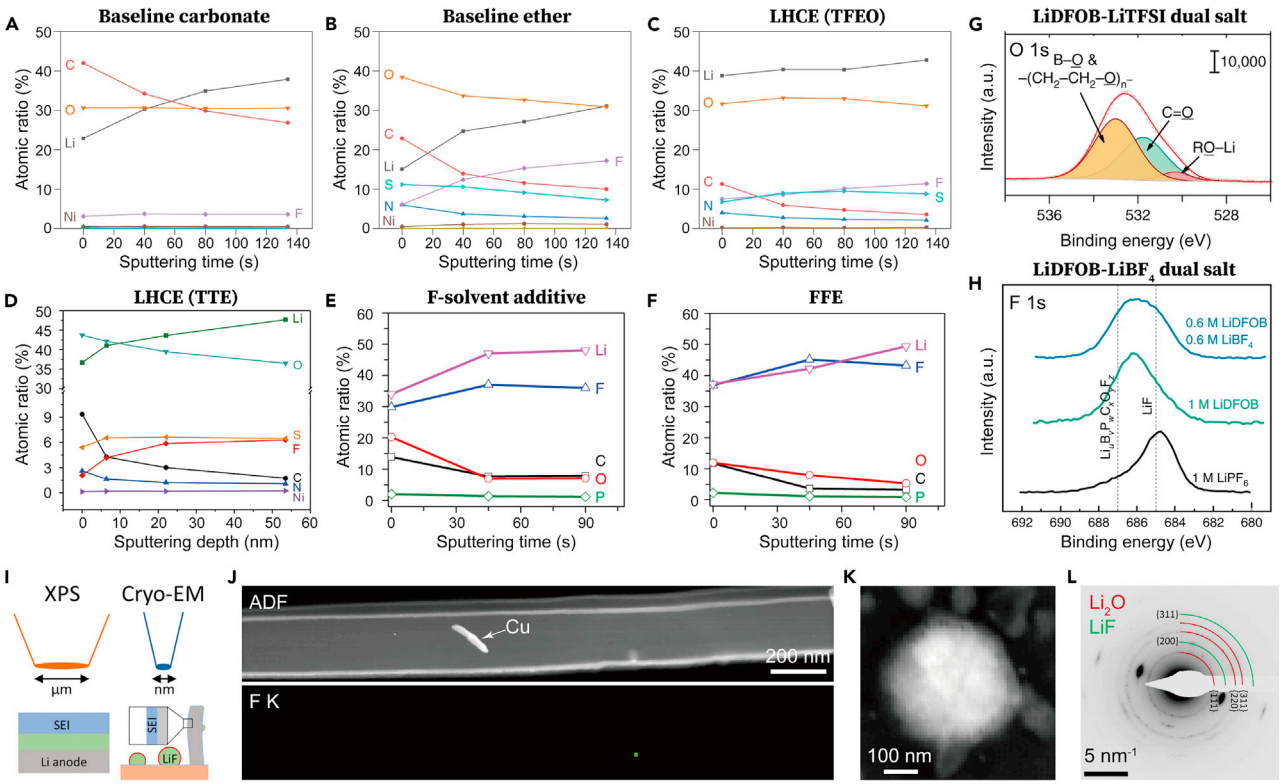


Figure 7. SEI chemistries

(A–F) XPS depth profile of the elemental ratio within the surfaces of Li deposited in: 1 M $LiPF_6$/EC-EMC-2%VC (A), 1 M LiFSI/DME (B), 1 M LiFSI/DME-TFEO (C), 1LiFSI-1.2DME-3TTE (D), 1 M $LiPF_6$/DME-FEC (E), 1 M $LiPF_6$/FEC-FEMC-D2 (F). (A–C) Reproduced with permission.[42] Copyright 2019, Nature Publishing Group. (D) Reproduced with permission.[50] Copyright 2019, Elsevier. (E and F) Reproduced with permission.[52] Copyright 2018, Nature Publishing Group.

(G) XPS O 1s spectra of Li deposited in 2 M LiTFSI-2 M LiDFOB/DME dual-salt electrolyte. Reproduced with permission.[68] Copyright 2018, Nature Publishing Group.

(H) XPS F1s spectra of Li deposited in a LiDFOB-$LiBF_4$/FEC-DEC dual-salt electrolyte. Reproduced with permission.[55] Copyright 2019, Nature Publishing Group.

(I) Schematic showing the difference between XPS and Cryo-EM characterizations on SEI chemistry.

(J) Cryo-STEM EELS mapping a Li filament with a single LiF particle.

(K) Cryo-STEM ADF image of LiF nanoparticle formed on the TEM grid electrode surface.

(L) SAED pattern of a LiF particle. LiF and Li_2O reflections are visible. (I–L) Reproduced with permission.[131,132] Copyright 2020 and 2021, American Chemical Society.

In contrast, more inorganic compositions in SEIs were observed within most advanced LMB electrolytes. For HCEs and LHCEs, the inorganic SEI chemistry was attributed to the anion-rich Li-ion solvation environments. For example, with 4 M LiFSI/DME,[34] researchers showed through theoretical modeling that only 3% of FSI^- anions were uncoordinated and only 6% of Li-ions were solely solvated by DME. This led to an inorganic-rich SEI dominated by anion decomposition, instead of a SEI that was rich in ether reduction products, such as ROLi.[34] Such a SEI chemistry also appeared in LHCEs, because the incorporation of diluents barely altered the Li-ion solvation environments. In 1 M LiFSI/DME-TFEO[42] and 1LiFSI-1.2DME-3TTE,[50] the SEIs were observed to be rich in $–SO_2F$ and $–SO_x$ functional groups and other inorganic components, such as LiF, Li_2S, and Li_3N. Carbon ratio was shown to be below 10% in both electrolytes (Figures 7C and 7D), confirming the inorganic nature of SEIs. Similar SEI chemistry was also observed for 1 M LiFSI/FDMB,[35] where the average anion/solvent ratio increased to ~3.29 compared with ~2.31 in 1 M LiFSI/DME. In these cases, the anion-rich Li-ion solvation

environments originate from the incorporation of weakly solvating molecules, which are usually fluorinated ethers. The electron-withdrawing effect of fluorine could weaken the coordinating capability of solvents' oxygen atoms toward Li-ions, leading to enhanced ion pairing. Besides this approach, another way to enable inorganic-rich SEIs is to use solvents with inorganic decomposition products. Fan et al. showed that the addition of FEC into baseline carbonate electrolyte could reduce the carbon ratio in SEIs to ~10% (Figure 7E),[52] together with significantly enhanced fluorine species originated from FEC decomposition. By completely replacing the baseline carbonates with fluorinated solvents, carbon ratio in the SEI was further reduced to below 10% (Figure 7F). Overall, inorganic-rich SEIs are observed in most advanced LMB electrolytes and considered critical for an improved LMA cycling performance.

Organic-rich SEIs

Organic-rich SEIs were rarely found to enhance LMA performances and mostly reported in dual-salt electrolytes with LiDFOB as the Li salt. One example was 2 M LiDFOB-2 M LiTFSI/DME. Through a series of characterizations, Jiao and co-workers[68] observed enrichment in $-(CH_2-CH_2-O)_n-$ (Figure 7G) and attributed this to the cross-linking or polymerization reactions induced by the electron-deficient state B atoms after B–O bond breakage. Another example was the dual-salt LiDFOB/$LiBF_4$ electrolyte reported by Dahn and co-workers.[55] XPS analysis of the F 1s fine structure showed that the SEI formed with $LiPF_6$ was predominantly composed of LiF, whereas that formed with LiDFOB mostly consisted of organic F species (Figure 7H). In the dual-salt LiDFOB/$LiBF_4$ electrolytes, equally strong signals were observed for LiF and organic F, which was considered responsible for its excellent performance in LMBs.

Where is LiF?

LiF is an almost universally observable SEI component in both conventional and advanced electrolytes and considered critical for Li metal cyclability.[52,112,116] In previous studies, LiF was predominantly observed via XPS F 1s peak assignments. However, a discrepancy on where LiF is located emerged when Cryo-EM results showed the absence of LiF within the compact SEIs in various electrolytes.[35,42,69,113,119] This discrepancy was resolved when the different spatial resolutions of XPS and Cryo-EM were taken into consideration (Figure 7I). The in-plane spatial resolution of XPS is limited to micrometer scale, meaning that the spectra represent signals from a μm-sized area with not only Li grains but also other particles dispersed on their surfaces. Through Cryo-EM and electron diffraction characterizations on a TEM grid electrode, Huang et al.[131,132] discovered that LiF deposited as crystalline nanoparticles with ~100–400 nm sizes on the current collector and, sparsely, on Li metal (Figures 7J–7L). These LiF nanoparticles were likely the origin of the LiF signals observed in the XPS characterizations. A similar morphology of LiF was also observed by Jurng et al.[96] Because LiF did not appear in compact SEIs that were in direct contact with Li metal, it was termed as an indirect SEI.[131,132] Consequently, LiF may not serve a crucial role in Li direct surface passivation as proposed previously.[52,112] However, the high surface energy and low Li adatom surface diffusion barrier[47,116] of LiF might still play a role in promoting horizontal Li deposition. Further studies directly correlating the spatial distribution of LiF to the morphology of electrodeposited Li deposition are required to fully reveal the role of LiF in Li metal cycling.

FUTURE DESIGN AND METRICS FOR ADVANCED LMB ELECTROLYTES

Although various strategies of electrolyte engineering have been reported before and summarized above, wide space still exists for the development of next-generation liquid

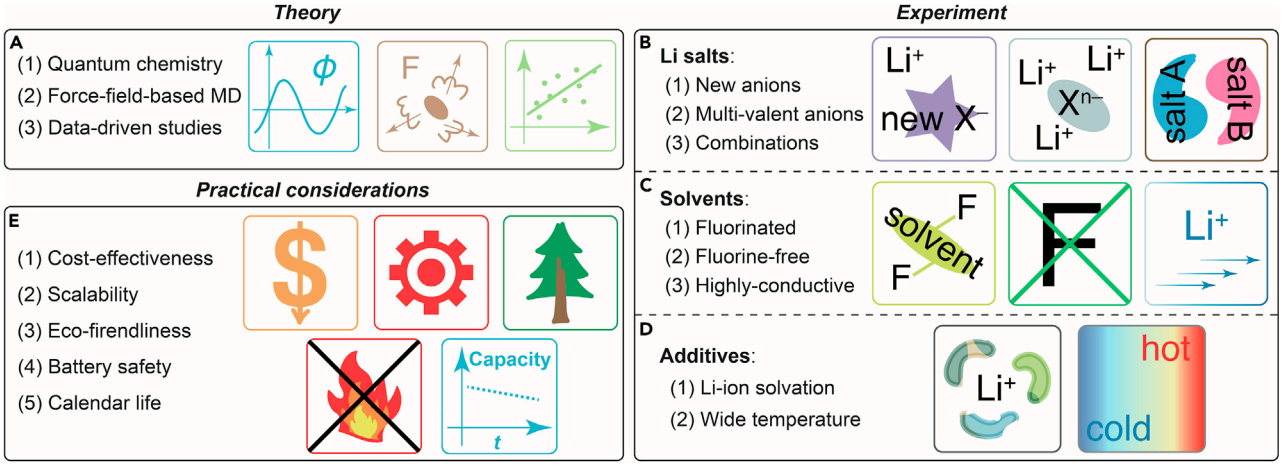


Figure 8. Future directions for LMB electrolyte research

(A) Design inspired by theoretical models: quantum chemistry, molecular dynamics and data-driven analysis.

(B–D) Experimental efforts on the electrolyte components, which are Li salts, solvents, and additives, under the guidance of theories.

(E) Important practical considerations beyond novel experimental advancements.

electrolytes. Both experimental and theoretical efforts are needed and, herein, we provide potential directions for exploring future electrolytes.

Theoretical simulations and data-driven studies

Theoretical simulations

A battery is a complex system composed of various components, including the anode, cathode, electrolyte, and SEI. The multiscale processes in batteries are further complicated by mass transport and side reactions.[133] All of these factors render it nontrivial, if not impossible, to design Li electrolytes following a few principles or using only one theoretical or simulation tool. Nevertheless, molecular simulations at different levels can provide essential insights about different processes in a battery.[134] Simulation should, therefore, be utilized as a critical tool in guiding the future development of advanced LMB electrolytes (Figure 8A).

Quantum chemistry calculations at different levels (semi-empirical methods, density functional theory, and wave function level) are usually adopted to gather single-molecule information of electrolyte components. Representative quantities include the HOMO, LUMO, ionization potential, and electron affinity of individual electrolyte components.[135] They provide key properties of electrolyte components without performing detailed and expensive condensed-phase simulations. These quantities from quantum calculations can give decent estimations of many important properties, such as electrochemical window, solubility of Li-ions, and reductive/oxidative stability. Therefore, high-throughput quantum calculations are a powerful method for screening candidate electrolytes from a pool of possible components.[136–139] Furthermore, quantum calculations based on clusters can also predict the interaction between Li-ions and other electrolyte components.[140] Infrared spectroscopy and NMR shifts from cluster calculations can be used to compare with and explain experimental data to gather more insights about the Li-ion solvation environments[141] in advanced electrolytes.

Force-field-based MD simulations can simulate electrolytes at larger scales of up to tens of nanometers, which makes the study of bulk electrolyte properties possible. The

solvation structure around each Li-ion can be obtained explicitly in these types of simulations.[141,142] Besides thermodynamic properties, MD simulations can also be used to study dynamic properties,[142,143] such as viscosity, dielectric relaxation, diffusivity, and ionic conductivity. Therefore, MD is a powerful tool for obtaining insights about the roles of different electrolyte components on key properties of advanced electrolytes. Additionally, MD simulations based on *ab initio* methods[144,145] or reactive force fields[146,147] can also shed light on the evolution of SEI. This makes it a promising method to reveal and rationalize the special interfacial properties enabled by advanced electrolytes, guiding further electrolyte formulation optimizations.

Data-driven studies

Recently, data-driven analysis and design have revealed great potential for battery developments.[139,148–151] Sendek et al. developed a machine learning (ML) model that predicted ionic conductivity of solid state electrolytes for LMBs based on quantum calculation data.[152] Attia et al. developed a model that can predict the cycle life of batteries from voltage profiles of the first 100 cycles[153] and further used the model to obtain the optimal fast-charging protocol through a closed-loop optimization process.[154] Despite the liquid electrolyte formulations developed for LMBs, there remains a vast and largely unexplored chemical design space, providing opportunities to use data-driven analysis. High-throughput screening of electrolytes through calculations, combined with data-driven approaches, can make electrolyte discovery more accessible (Figure 8A).

Two types of approaches can potentially be adopted to optimize the LMB electrolyte formulation. First, theoretical simulation data can be utilized to predict electrolyte properties through a bottom-up manner. Specifically, calculated properties, such as HOMO/LUMO quantities and solvation structures, can be fed into ML models, which can be subsequently used to guide electrolyte designs. On the other hand, a top-down approach can also be adopted. Experimental data, such as CE or full-cell cycle life, could be used to train a ML model to extract the key features (such as chemical compositions) of a certain electrolyte that determine its performance. If these approaches are successful, future experimental screening of the electrolyte can be accelerated.

Molecular designs for LMB electrolytes

Recent works on electrolyte engineering usually focused on tuning the recipe with commercialized chemicals. While this progress is undoubtably critical and worth exploring,[98,99,155,156] we believe that an important future direction is to develop rationally designed structures and to explore new chemicals. Organic and inorganic synthetic methodologies, therefore, can be the central tools in the next phase of electrolyte research. This section is categorized by common components in liquid electrolytes, i.e., Li salts, solvents, and additives, and an outlook is provided for each component.

Li salts

As the source of Li-ion transport, Li salts are critical for advanced electrolytes.[157] Three potential directions for developing Li salts are (1) new-anion salts, (2) multivalent anions, and (3) combination of different salts (Figure 8B).

The anion of the Li salt determines the quality of SEI and, as a result, the CE. Therefore, opportunities exist for designs of Li salts based on novel anions to fulfill the urgent demand for higher Li efficiency. For example, promising candidates include recently rising Hückel-type anions, such as 4,5-dicyano-2-(trifluoromethyl)imidazole (TDI) and 4,5-dicyano-2-pentafluoroethylimidazolate (PDI),[158] fluorinated

phosphates, such as difluorophosphate (DFP),[159,160] $[PF_3(CF_2CF_3)_3]^-$ (FAP),[161] tetrafluoro oxalato phosphate (TFOP),[162] various fluorosulfonyl imides,[163–167] new borates,[168–170] acetates,[171] and even fluorine-free anions, such as $AlCl_4^-$,[172] and cyano-based anions.[158,173,174]

Despite these novel yet monovalent anions, one can further create multivalent anions for Li salts to enhance the Li-ion density and salt utilization by reducing the anion/cation ratio. For example, several divalent fluorosulfonyl imide-based salts were proposed and synthesized.[175,176] It is worth noting, however, that multivalent anions may also introduce poor solubility; therefore, fine tuning of the size and substitution of multivalent anions can be applied to enable decent salt dissolution. Another potential issue to be considered is the strong electrostatic interaction between multivalent anions and Li cations, which may negatively affect the transport properties.

Besides the design and synthetic efforts mentioned above, perfect recipes for different Li salts (i.e., multi-salt approach as elaborated above) can be a supplemental approach to synergically boost Li metal performance. Combinations of both existing salts with synthesized salts can be pursued. This point is even more critical if the synthetic salts are less cost-effective or scalable than commercially available ones.

Electrolyte solvents

Physicochemical properties of electrolyte solvents are crucial to LMBs.[26,177] The solvent should also be stable toward cathodes, but, herein, we mainly focus on the general design directions for electrolyte solvents that are stable toward LMAs: (1) new fluorinated solvents, (2) fluorine-free eco-friendly solvents, (3) highly conductive solvents, (4) non-flammable/safe solvents, and (5) combination of different solvents (Figure 8C).

Fluorine is a well-known "magic" element to stabilize LMBs.[178,179] Although a number of fluorinated solvents and diluents have been reported (e.g., fluoro-carbonates,[52,177,180,181] fluoro-esters,[177,182] fluoro-acetamide,[183] fluoro-sulfonamide,[53,67] fluoro-ethers,[32,35,42,184,185] fluoro-phosphates,[186–188] fluoro-sulfones,[189,190] fluoroalkanes,[121] fluorinated aromatic diluents,[191,192] fluorinated ionic liquids,[193] etc.), few newly synthesized compounds have been applied to LMBs. Opportunities exist to take the advantage of organic synthetic tools to explore novel small molecules as electrolyte solvents or diluents. Another direction in need is deeper understandings of the structure-property relationships between fluorination sites on the molecules and impact on battery performance. Particularly, the effect of electrostatic/non-covalent Li–O and/or Li–F interactions on solvation structure and overall performance deserves further exploration.

Besides fluorinated solvents, an opposite yet promising approach is fluorine-free solvents. Eliminating fluorine atoms in solvent molecules will have profound impacts for eco-friendliness.[173,174,194] The challenge is to maintain or even enhance the battery performance while removing fluorine substitutions. One promising and recently explored direction is to construct steric molecules to enable weakly solvating solvents for better Li metal stability.[62,91,195]

Solvents with great salt solubility and high ionic conductivity are essential for fast-charging capabilities and low-temperature operation of LMBs. Special solvation environments tailored by the electrolyte design can modulate the charge transfer process to enable low-temperature cyclability.[196] Nevertheless, the molecular-level design principles of solvent structures remain elusive in this field. One may

learn from the electrolyte design for fast charging and low-temperature cycling of LIBs,[197] and modify the solvents, such as esters and nitriles, as well as ethers, according to the requirements of LMBs.

Similar to Li salts, solvents can also be mixed for an ideal formulation. Combinations of the fluorinated solvents, diluents, fluorine-free solvents, and highly conductive and nonflammable solvents (as will be discussed in the later Battery safety section) may yield synergistic effects and improve the CE of Li metal.

Additives

Additives make up small proportions of the electrolyte, but they can dramatically affect the performance of LMAs.[100] Countless additives have been reported, and further development should expand beyond the salts and solvents mentioned above. While the functioning path of an additive may vary, its most important roles are (1) tuning SEI and Li^+ solvation structure for better Li metal CE, and (2) enabling a wide range of temperatures for battery operation (Figure 8D).

Additives can tune the SEI in several ways[198]: (1) an additive itself decomposes on Li metal and forms stable products in SEI, (2) an additive modifies the solvation structure and renders preferential decomposition of other components to form a stable SEI, or (3) synergistic function of (1) and (2). Regardless of the detailed mechanism, the target is to create a more stable SEI on LMAs. Systematic studies on the structure-property relations between additive structure and SEI quality is lacking, and upgrading existing additives, such as $LiNO_3$, is an urgent direction. Research into the effect of different dissolution promoters on cycling efficiency are needed, which will benefit further design of novel promoter structures. In the meantime, other novel additives to further improve SEI quality and boost Li metal performance can be designed and obtained through cost-effective and large-scale synthesis. Additives for wide temperature operation will be discussed in the later section.

Practical considerations for LMB electrolytes

In addition to the above materials and chemistry-based outlooks, we need to pay attention to the practicality of electrolyte strategies from the perspectives of industry, consumers, and market. Detailed categories and elaborations are as follows.

Cost-effectiveness, scalability, and eco-friendliness

The manufacturing cost is a critical consideration for the commercialization of LMBs, so designs of new Li salts, solvents, and additives must consider the cost-effectiveness of an electrolyte. The price of starting reagents, the step length of synthetic procedures, the cost of reaction conditions and purification (e.g., strict and harsh conditions increase the cost), and the yield of products (Figure 8E) are important components of the cost and determine the intrinsic scalability of the electrolyte. Electrolytes account for ~15% cost of the whole Li-ion battery, and it is safe to expect higher cost for electrolytes in LMBs due to the use of more expensive Li salts and newly synthesized solvents or additives.[26,157,199] It is noteworthy that the mass production of conventional carbonate electrolytes over the pass decades significantly optimized the synthetic pathways so that the cost could be reduced. This means that recent advanced electrolytes do have significant space for cost reduction if put into mass production, and this still reminds us to prioritize cost-effectiveness during electrolyte designing.

Moreover, green chemistry and atom economy should be applied to mitigate or even eliminate the pollution generated during battery chemical manufacturing.[26] For example, organic-solvent-free method such as ball milling can be implemented

in salt synthesis,[159] whereas organic synthetic procedure can be designed to generate fewer side products. All these require the manufacturing and engineering efforts from chemical industry.

Battery safety

Safety is another critical concern in realistic LMBs for consumers, so non-flammable solvents and/or additives with optimized battery performance are worth developing. Generally, phosphates, sulfones, and fluorine-rich solvents are fire retardant, but they often have poor Li metal CE.[52,66,85,186,200] Post-modification of these chemicals can be conducted to improve their cycling performance while retaining the non-flammability. The incorporation of special additives containing halogen or phosphor elements can mitigate flammability as well.[100] It is noteworthy that battery safety is complex and should be examined within commercial level cells under realistic testing conditions including nailing tests or heating,[39,201] because thermal runaway is related to factors beyond solvent flammability, such as exothermic reactions between electrolyte components and electrodes and the subsequent gassing issue.[39,201] In a recent report,[201] Ouyang et al. found that even the HCE based on fire-extinguishing solvent, trimethyl phosphate (TMP), would catch fire under a nailing test, although it had been assumed non-flammable for long. This new learning calls for more practical and careful examination for LMB safety when using advanced electrolytes.

Calendar life

Recent developments of LMB electrolytes particularly emphasize the improvement in battery cycle life, whereas they place less attention on the cell calendar life (capacity retention after storage at open circuit), which is equally critical for the commercialization of LMBs. Although advanced LMB electrolytes have dramatically enhanced the cycling efficiency, full-cell calendar life is still not satisfying.[202,203] Specifically, Boyle et al. found that advanced electrolytes that enabled long LMB cycle life demonstrated similar calendaring aging rates as conventional electrolytes.[202] This indicates that enhancement of cycle and calendar life follows distinct mechanisms. High cycling efficiencies can be accomplished through the promotion of large Li deposition granules that minimizes "dead Li" formation.[37] During calendar aging, however, thickening of SEIs and even the formation of extended SEIs were observed for various electrolytes and considered as the dominant source of capacity loss.[202] Therefore, achieving satisfying calendar life requires more emphasis on the passivation of Li with SEI. Consequently, future advanced LMB electrolytes are required to not only enable low surface area of deposited Li, but also promote highly passivating SEIs with stable chemical components and dense microstructures that are non-permeable to electrolytes. One possible direction that has been found is to introduce $LiNO_3$ additive that effectively passivated the Li surface during calendar aging.[202] The designs for long calendar life electrolytes may be enabled in the future by further and deeper understandings on Li-ion solvation and the corresponding SEI formation pathways. Moreover, the calendar life needs to be verified with full-cell configurations under industry standards, because cathodes and realistic cycling conditions matter as well in calendar aging.

Wide temperature range operation

Another important yet less-studied area is the electrolytes to enable wide temperature operation of LMBs, which can be achieved through tuning of SEI and Li-ion solvation structure. Generally, low-temperature cycling requires highly conductive SEIs and lower de-solvation barrier,[18,90,182] whereas high-temperature performance depends more on the robustness of SEI and CEI.[113] The latter direction is less studied in LMB field, although several reports on LIBs showed unsaturated bonds on phosphate[204] or silane[205] structures helped with cycling at elevated temperatures.

Some novel salts[113,168] and a conventional electrolyte 1 M LiTFSI/DOL-DME with $LiNO_3$ additive[113] have been found to enable better cycling of high-temperature LMBs, but the molecular design methodology for high-temperature LMB additives is missing. On the other hand, the low-temperature operation of LMBs is attracting research attention due to the demand of running electric vehicles in cold weather. Several examples such as liquified gas electrolytes,[120] weakly solvating solvent[196] and all-fluorinated solvents[90,182] have been demonstrated to enable stable cycling of LMBs under extremely low temperature up to −95°C; however, some of them suffer from low boiling point, limiting their practical applications at room temperature. Therefore, systematic studies and more molecular designs are still needed for future low-temperature electrolytes.

Critical metrics for evaluating developed electrolytes

It is critical for the community to agree on a set of standardized criteria for evaluating the performance of electrolytes. Several journals announced the checklists[206,207] for experimental battery research and these can be good practices to implement into electrolyte works. We suggest that critical cell conditions need to be clearly and directly stated in the figure plots, figure captions, tables, or methods section. These critical parameters include: (1) Li metal thickness or Li inventory amount, (2) cathode type, loading, active material/conductive carbon/binder ratio, (3) electrolyte amount, i.e., E/C ratio, (4) cycling conditions such as charge/discharge current, voltage cut-off range, temperature, external pressure, etc., (5) number of cycles at 80% capacity retention, and (6) achieved specific capacity based on cathode materials if coin cells are tested or specific energy in pouch cell cases. Beyond these key parameters, we strongly recommend that electrolyte works (or even all battery works) show at least two replicated cell data for any single cycling condition.

Besides, special recommendations for measuring and reporting CE need to be considered. It is noteworthy at the beginning that the correlation between CE and full-cell cycling performance may not be as tight as we previously expected, because a small excess of Li inventory in the testing cell can yield overestimation of CE.[38] However, the CE of either LMAs or full cells is still an important and direct indicator for Li metal cycling performance, and, therefore, is recommended to be reported following certain protocols. Most previous reports plotted CE with y axis ranging from 0% to 100%, which disabled unambiguous observation of either initial CE activation[35] or its fluctuation. For Li||Cu half cells, three plots are suggested to clearly report Li metal CE: (1) zoomed-in plot starting from the first cycle to the initial tens of cycles, to show how fast Li metal anode can be activated, (2) long-term cycling CE with y axis scale at least from 90% to 100%, to show the efficiency as well as fluctuation over cycles, and (3) modified Aurbach protocol,[38,208] which determines the average Li metal CE. For Li||cathode full cells or Cu||cathode anode-free cells, (1) and (2) can be implemented in the report.

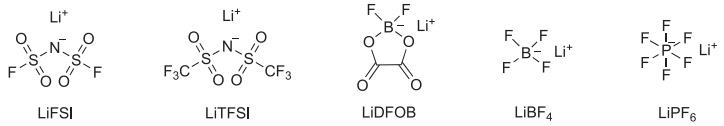
Because the testing protocol will vastly influence the CE results, we, herein, summarize and propose general protocols for standardization:

(1) Li||Cu half cells: the standard and commonly accepted current density and areal capacity are 0.5 mA/cm^2 and 1 mAh/cm^2, respectively, which can be the minimum required values for Li||Cu long-term plating/stripping and modified Aurbach protocol.[38,208] Higher values are more desired.
(2) Li||cathode and Cu||cathode full cells[38]: high-loading (up to 4–5 mAh/cm^2), high-voltage (>4.2 V) cathodes are desired for maximizing energy density; however, LFP should also be seriously considered due to its low cost and recent prevalence in market. Generally, using a cathode with areal loading >1 mAh/cm^2 in liquid

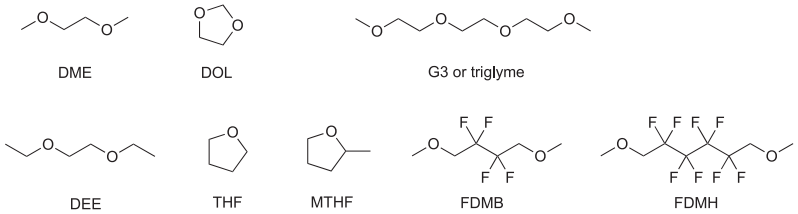
electrolyte works is highly recommended. Current density of >0.5 mA/cm^2 is desired for low-loading (<2 mAh/cm^2) full cells; while for high-loading (4–5 mAh/cm^2) ones, slower cycling rate (e.g., C/10 charge C/3 discharge) can be implemented.

APPENDIX 1: KEY CHEMICAL STRUCTURES OF LI SALTS, SOLVENTS, ADDITIVES, AND DILUENTS IN TABLE 1 AND 2.

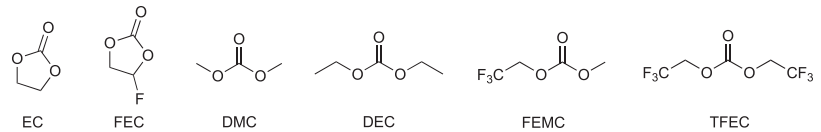
Lithium salts



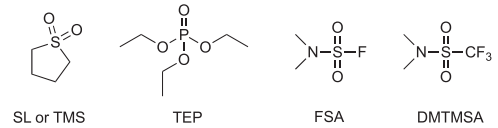
Ether solvents



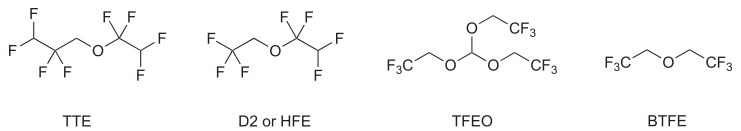
Carbonate solvents



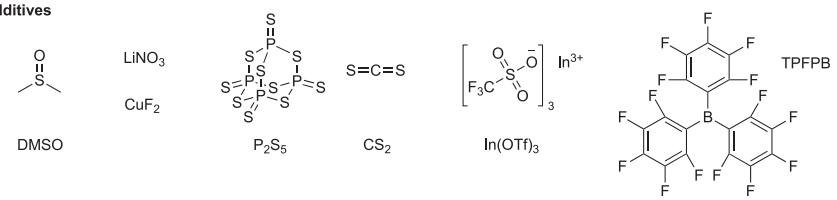
Other solvents



Diluents



Additives



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DECLARATION OF INTERESTS

The authors declare no competing interests.

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